DIGITAL DICE USING LOGIC GATES

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Digital dice are simple electronic systems that generate and display numbers from **1 to 6**, similar to a conventional dice used in games. They demonstrate the practical application of digital electronics, logic gates, counters, flip-flops, and LED-based display circuits. Digital dice systems provide a simple and reliable alternative to mechanical dice and can be implemented using basic digital logic components.

This project presents the design and implementation of a **Digital Dice Using Logic Gates**. The system generates a sequence of digital states corresponding to the numbers 1 to 6 using a clock signal and digital counting circuitry. Logic gates are used to control and decode the generated states, while LEDs arranged in a dice pattern are used to display the corresponding number.

When the circuit is activated, the counter continuously changes its state, and a user-controlled action can be used to select one of the six possible outputs. The selected number is displayed through the appropriate LED pattern, representing the face of a conventional dice. The project demonstrates fundamental concepts such as **digital counting, logic gate operation, state generation, decoding, and digital display techniques**. It provides a simple and practical application of digital logic circuits in an interactive electronic system.

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LIST OF ABBREVIATIONS

ABBREVIATION	MEANING
AC	Alternating Current
DC	Direct Current
IC	Integrated Circuit
IoT	Internet of Things
LED	Light Emitting Diode
LFSR	Linear Feedback Shift Register
VLSI	Very Large Scale Integration
Q_0	Counter Output Bit 0
Q_1	Counter Output Bit 1
Q_2	Counter Output Bit 2

CHAPTER 1
INTRODUCTION OF DIGITAL DICE

1.1 INTRODUCTION

Digital electronics is an important field of electronics engineering that deals with the processing and representation of information using discrete signal levels, generally represented by binary values such as 0 and 1. Digital systems are widely used in computers, communication systems, industrial automation, consumer electronics, control systems, and entertainment applications. Logic gates, flip-flops, counters, registers, and combinational and sequential circuits form the basic building blocks of digital electronic systems.

A dice is a small object traditionally used in games to generate a number between 1 and 6. Conventional dice are generally made of plastic or other solid materials and produce an outcome based on their physical motion. Although mechanical dice are simple to use, their operation depends on physical movement and cannot easily be integrated with electronic systems. A digital dice provides an electronic method of generating and displaying numbers from 1 to 6.

The **Digital Dice Using Logic Gates** is a simple digital electronic system designed to produce numbers from 1 to 6 and display the selected number using LEDs arranged in the form of a conventional dice. The system demonstrates how basic digital logic can be used to create a practical application. It combines a clock or pulse signal, counting circuitry, logic gates, and an LED display section to generate and represent the dice numbers.

The basic operation of the digital dice is based on sequential generation of digital states. A clock signal is applied to a counter circuit, causing its output to change from one state to another. The required states corresponding to the numbers 1 to 6 are selected using appropriate logic circuitry. When the counter reaches the required state, the corresponding LED pattern is activated.

For example, when the generated state represents the number **1**, the centre LED of the dice pattern is switched ON. For the number **2**, two LEDs positioned at opposite corners are activated. Similarly, different combinations of LEDs are used to represent numbers 3, 4, 5, and 6. This makes the output easy to understand and visually similar to a real dice.

The project mainly focuses on understanding the practical application of **logic gates and digital counting circuits**. It provides an opportunity to study the relationship between binary states and visible digital outputs.

The circuit can be designed and tested using simulation software such as Logisim, Proteus, or similar digital circuit simulation platforms before practical implementation.The project is particularly useful for students learning digital electronics because the circuit is relatively simple while demonstrating several important concepts. It provides practical knowledge of logic gate operation, binary counting, state generation, decoding, LED interfacing, and digital display techniques.

Thus, the Digital Dice Using Logic Gates serves as a simple but effective example of how fundamental digital electronic components can be combined to develop an interactive electronic system.

1.2 Objective

The main objective of this project is to **design and implement a Digital Dice using basic digital logic circuits and LEDs**. The system is intended to generate numbers from 1 to 6 electronically and display them in a dice-like pattern.

The major objectives of the project are as follows:

- **To understand the fundamentals of digital electronics** and the operation of basic logic gates.
- **To design a digital number generation circuit** capable of producing six valid states corresponding to the numbers 1 to 6.
- **To implement a counting mechanism** that continuously changes the digital state using a clock or pulse signal.
- **To use logic gates for decoding and controlling the output**, so that the appropriate LED combination is activated for each generated number.
- **To design an LED-based dice display** in which the LEDs are arranged similarly to the dots on a conventional dice.
- **To understand the relationship between binary states and decimal outputs** through practical implementation.
- **To study the operation of a Mod-6 counting sequence**, where the valid output states are restricted to six states.
- **To provide a simple electronic alternative to a conventional mechanical dice** for demonstration and educational purposes.
- **To simulate and verify the circuit operation** using digital circuit simulation software before practical implementation.
- **To develop practical knowledge of combinational and sequential digital circuits** through the construction of a small-scale electronic project.
- **To improve understanding of LED interfacing**, including the control of individual LEDs

according to the generated digital state.

- **To demonstrate the practical application of digital logic** in an interactive system.

The project also aims to provide a foundation for developing more advanced electronic dice systems in the future.

The basic design can be enhanced by incorporating improved random-number generation, multiple dice, seven-segment displays, microcontrollers, wireless communication, or electronic gaming applications.

1.3 Scope

Digital dice using logic gates has a wide scope in digital electronics and educational applications. In many electronic systems, digital circuits are used to generate, process, and display information in a simple and reliable manner. A digital dice is a basic application that demonstrates how logic gates and counting circuits can be combined to generate numbers from 1 to 6.

The scope of this project includes the design and implementation of a digital dice using basic logic gates, a counting circuit, and LEDs for displaying the output. The project demonstrates how digital states can be generated and decoded to produce different dice numbers. It also provides knowledge about clock signals, counters, logic operations, and LED-based display circuits.

This project can be further expanded by integrating additional features such as improved random number generation, dual dice operation, seven-segment display, and push-button control. The system can also be modified using microcontrollers or other digital devices for more advanced applications.

The concepts used in this project are applicable in various fields such as educational demonstrations, electronic games, digital learning systems, probability experiments, and interactive entertainment devices. Therefore, the study of digital dice using logic gates provides a simple and practical understanding of digital circuit design and implementation.

1.4 Existing System

In the existing system, conventional mechanical dice are commonly used for generating numbers between 1 and 6. These dice are generally made in a cube shape with different numbers of dots marked on each face. The dice is physically rolled by the user, and the number appearing on the upper face is taken as the result.

Traditional dice systems depend completely on mechanical movement and physical interaction. The result is obtained only after the dice is thrown and comes to rest. Since the output is represented physically using dots, it cannot directly provide an electrical or digital signal to other electronic circuits.

Another limitation of conventional dice is that they are not directly compatible with digital systems and automated applications. Integrating a mechanical dice with an electronic system would require additional sensors or detection mechanisms to identify the resulting number.

Due to these limitations, conventional dice are less suitable for applications involving digital electronics, automation, and electronic gaming systems.

Therefore, a digital dice based on logic gates provides a simple electronic alternative that can generate and display the required numbers using digital circuits and LEDs.

1.5 Proposed System

The proposed system uses a digital logic circuit to generate and display numbers from **1 to 6** electronically. In this system, a clock or pulse signal is used to drive a counting circuit, which generates a sequence of digital states corresponding to the dice numbers.

The output signals from the counter are given to a **logic gate and decoding circuit**, which determines the required LED combination for each generated number. The logic gates process the binary signals and control the LEDs arranged in the form of a conventional dice pattern.

By activating the appropriate LEDs according to the generated digital state, the system displays the numbers **1, 2, 3, 4, 5, and 6**. The counter operates in a Mod-6 sequence, so after reaching 6, it returns to 1 and continues the sequence. A push-button or suitable control mechanism can be used to select the required output.

The proposed system provides several advantages such as **simple digital operation, clear LED display, low-cost implementation, and easy circuit design**. It eliminates the need for physically rolling a conventional dice and demonstrates the practical use of logic gates, counters, and digital display techniques.

This system is mainly useful for **digital electronics education, laboratory demonstrations, electronic games, and probability-based experiments**. The design can also be further enhanced by incorporating improved random number generation, dual dice operation, seven-segment displays, or microcontroller-based control.

CHAPTER 2

LITERATURE REVIEW

2.1 Literature Survey

The literature review provides an overview of the research and developments related to **digital dice, digital logic circuits, counters, and random number generation**. It helps in understanding the different techniques and methods that have been proposed for generating and displaying digital numbers. Many researchers and authors have studied digital logic circuits because of their importance in electronic systems, automation, gaming applications, and educational systems. The use of logic gates and counters provides a simple method for generating sequential digital states and controlling output displays.

Several studies have also focused on the design of **electronic dice systems** using counters, flip-flops, logic gates, and display circuits. These developments have helped in creating compact and reliable electronic systems that can replace conventional mechanical dice. The use of LED displays and digital decoding techniques also provides a simple and effective method for representing the generated numbers.

These studies provide valuable insights into the design and implementation of digital dice systems using basic digital electronics. The following table summarizes the contributions of various authors in the field of **digital logic, counters, random number generation, and electronic dice systems**.

S.NO	Authors	Title	Abstract	Disadvantages
1	Thomas L. Floyd	**Digital Fundamentals and Logic Circuits**	The author explains the fundamental concepts of digital electronics, including logic gates, Boolean algebra, combinational circuits, sequential circuits, and digital counters. These concepts provide the basic knowledge required for designing a digital dice circuit.	Requires a basic understanding of Boolean algebra and digital circuit concepts. Practical implementation may require additional hardware components.
2	R. P. Jain	**Modern Digital Electronics**	The author discusses digital logic design, flip-flops, counters, registers, and combinational circuits. The study provides useful information about counter-based state generation and logic circuits that can be applied to digital dice design.	Circuit design can become complex when a larger number of states or additional features are included. Hardware implementation requires proper component selection.

3	M. Morris Mano, Michael D. Ciletti	**Digital Design**	Their work explains Boolean logic, logic gate implementation, sequential circuits, counters, and state machines. These concepts are useful for designing a digital system that generates a sequence of states and decodes them into meaningful outputs.	Detailed digital design techniques may be complex for beginners. Advanced designs may require more hardware and careful state analysis.
4	Stephen Brown, Zvonko Vranesic	**Fundamentals of Digital Logic with Verilog Design**	The authors describe digital logic design and the implementation of combinational and sequential circuits. Their work provides knowledge about designing digital systems, state transitions, and output decoding techniques applicable to electronic dice systems.	Simulation and hardware description techniques may require additional software knowledge. Complex designs require more detailed verification.
5	A. Anand Kumar	**Fundamentals of Digital Circuits**	The author explains the operation of basic logic gates, flip-flops, counters, and digital display circuits. These concepts are directly useful for understanding how a counter can generate digital states and how logic gates can control an LED-based dice display.	Basic counter-based systems produce a repeating sequence and may not provide true random behavior without additional randomization techniques.

Table 2.1 Literature Survey

Summary of Literature Survey :

The literature survey shows that **logic gates, flip-flops, counters, and decoding circuits** are important building blocks for digital number-generation systems. The studies provide the theoretical foundation required to design a digital dice using simple digital circuits. Counter circuits can be used to generate sequential states, while logic gates can decode these states and control the corresponding LED patterns.

Based on the reviewed concepts, the proposed **Digital Dice Using Logic Gates** combines a clock signal, counter circuit, logic-gate decoding, and LED display to generate and represent numbers from 1 to 6. The project provides a simple practical implementation of the digital logic concepts discussed in the literature.

CHAPTER 3

CORE SYSTEM & HARDWARE COMPONENTS

3.1 Digital Dice System

A digital dice is an electronic system designed to generate and display numbers from **1 to 6**, similar to a conventional dice used in games. The system uses basic digital electronic components such as a clock or pulse generator, counter circuit, logic gates and LEDs. The generated digital states are decoded using logic gates and converted into suitable LED patterns to represent the numbers on a dice.

The main purpose of the system is to demonstrate the practical application of digital logic circuits in a simple and interactive electronic system. Instead of physically rolling a mechanical dice, the electronic circuit generates a sequence of numbers and displays the selected result using LEDs.

The digital dice consists of three major functional sections: the **clock/pulse generation section, counter and logic section, and LED display section**. The clock produces pulses that drive the counter. The counter generates the required digital states, while the logic circuit decodes these states. Finally, the LED display produces the corresponding dice pattern.

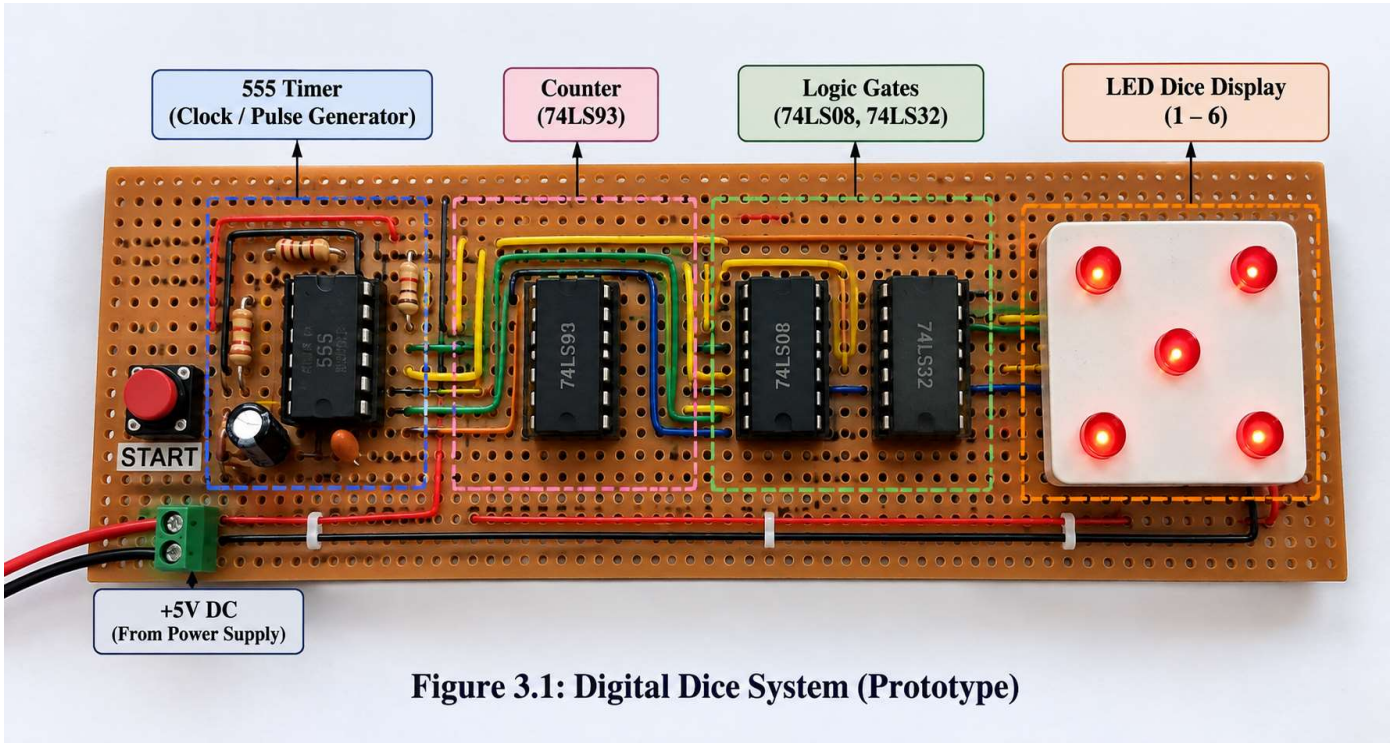


Figure 3.1: Digital Dice System (Prototype)

3.2 System Block Diagram

The digital dice system can be divided into three main blocks: **Clock/Pulse Generator, Counter and Logic Circuit, and LED Display Unit**. Each block performs a specific function and together they produce the required dice output.

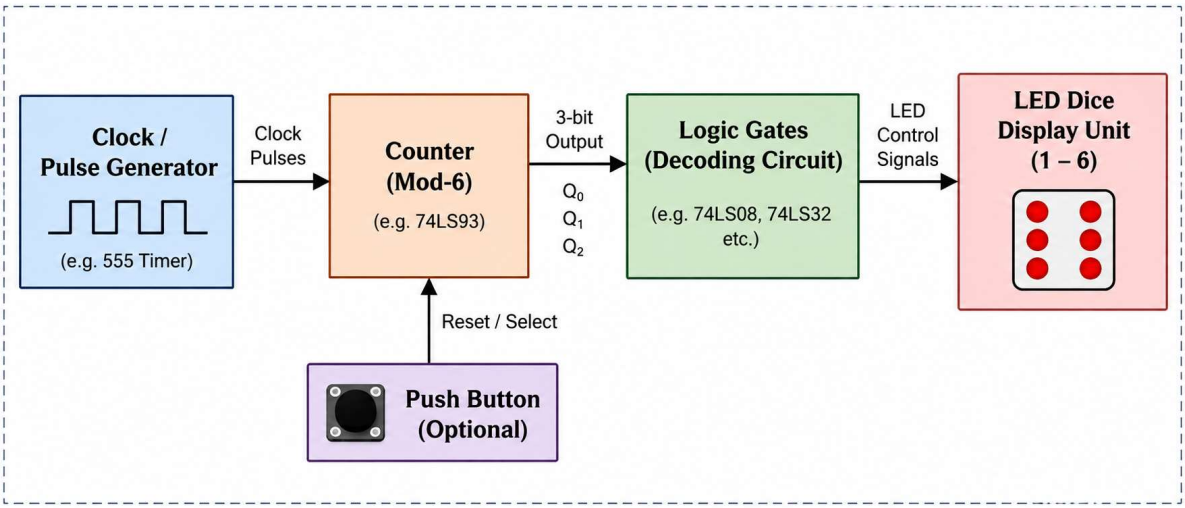


Figure 3.2: Block Diagram of Digital Dice

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3.2.1 Clock/Pulse Generator

The clock or pulse generator provides the timing signal required for the operation of the digital dice circuit. A digital counter changes its output state whenever it receives a clock pulse. Therefore, the clock signal determines how quickly the numbers generated by the counter change.

A suitable pulse generator can be implemented using a timer circuit or another digital oscillator. The generated pulses are applied to the counter as its clock input. When the clock frequency is sufficiently high, the counter continuously changes between its different states.

In the digital dice system, the clock section provides a continuous sequence of pulses so that the displayed number changes rapidly before the user selects the result

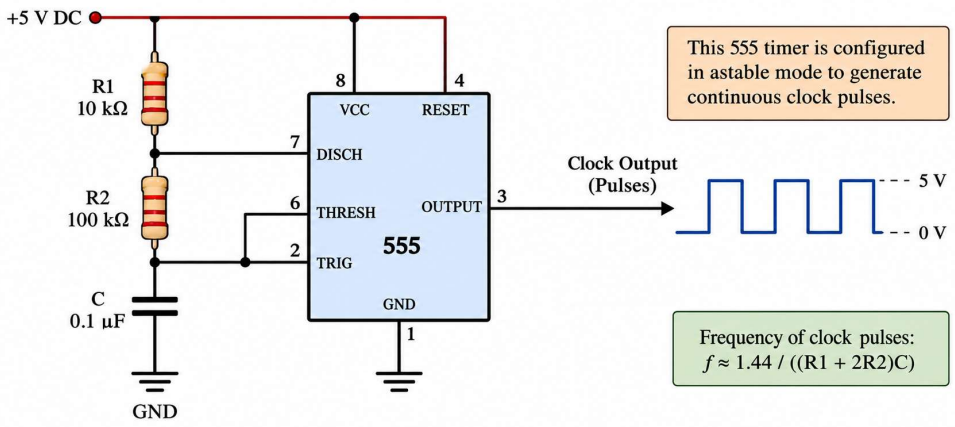


Figure 3.3: Clock/Pulse Generator using 555 Timer

3.2.2 Counter and Logic Circuit

The counter and logic circuit forms the main processing section of the digital dice. The counter receives clock pulses and generates a sequence of digital states. Since a dice has only six possible numbers, the counter is designed to operate as a **Mod-6 counter**.

The counter produces the sequence corresponding to numbers 1, 2, 3, 4, 5 and 6 and then returns to the first state. Logic gates are connected to the counter outputs to decode these states and generate suitable signals for controlling the LEDs.

The logic circuit therefore acts as an interface between the counter and the LED display. Each counter state corresponds to one particular dice number.

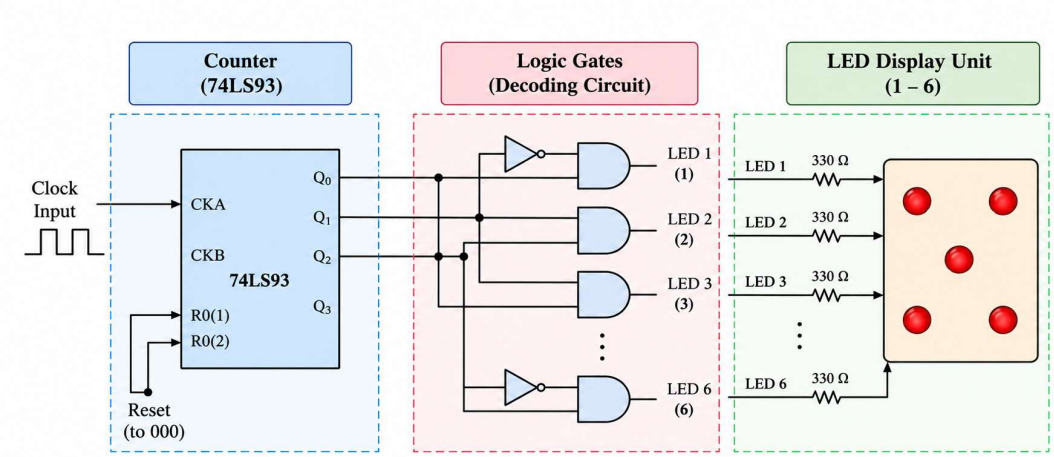


Figure 3.4: Counter and Logic Circuit

3.2.3 LED Display Unit

The LED display unit is the output section of the digital dice. LEDs are arranged in a pattern similar to the dots or pips on a conventional dice. Depending on the output received from the logic circuit, the required LEDs are switched ON.

For example, when the output corresponds to number 3, three LEDs are illuminated in a diagonal arrangement. Similarly, six LEDs are illuminated when the generated number is 6.

The LED display makes the output easy to understand because the user can directly observe the dice pattern without requiring a numerical display.

Table 3.1: Logic Gates Used in Digital Dice

Logic Gate	Function	Application in Digital Dice
AND	Produces HIGH when all inputs are HIGH	State detection
OR	Produces HIGH when any input is HIGH	Combining logic signals
NOT	Produces the complement of input	Inverting counter outputs

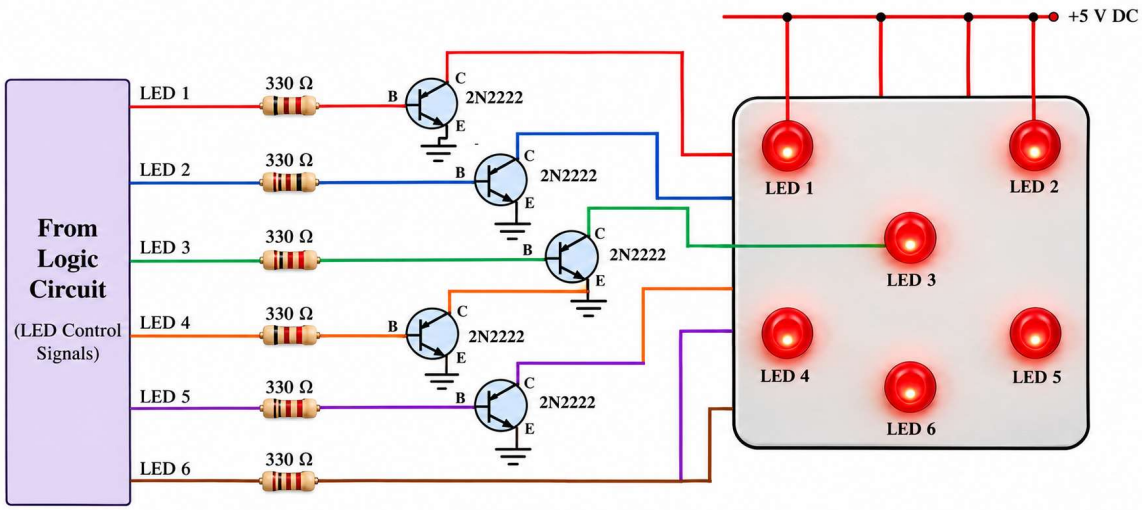


Figure 3.5: LED Dice Display Unit

3.3 Logic Gates Used

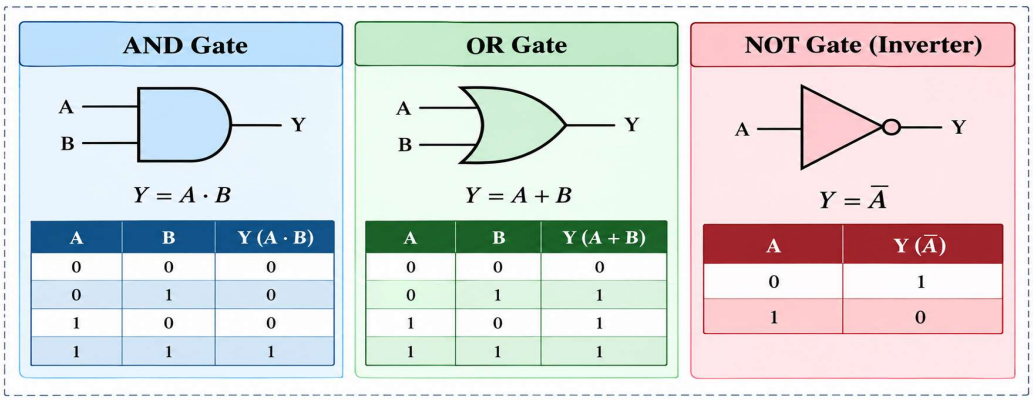
Logic gates are fundamental building blocks of digital electronic circuits. In the digital dice system, logic gates are used to decode the outputs of the counter and generate the required control signals for the LEDs.

The main gates used in the circuit can include **AND, OR and NOT gates**. The AND gate is used to detect a particular combination of counter outputs. OR gates can combine signals from different logic conditions, while NOT gates provide the required complemented signals.

The combination of these gates allows the circuit to identify each counter state and activate the corresponding LED pattern.

Table 3.1: Logic Gates Used in Digital Dice

Logic Gate	Function	Application in Digital Dice
AND	Produces HIGH when all inputs are HIGH	State detection
OR	Produces HIGH when any input is HIGH	Combining logic signals
NOT	Produces the complement of input	Inverting counter outputs



A	B	Y (A · B)
0	0	0
0	1	0
1	0	0
1	1	1

A	B	Y (A + B)
0	0	0
0	1	1
1	0	1
1	1	1

A	Y ($\overline{A}$)
0	1
1	0

Figure 3.6: Basic Logic Gate Symbols and Truth Tables

3.4 Counter Circuit

A counter is a sequential digital circuit that changes its output according to the applied clock pulses. It is commonly constructed using flip-flops and is used for counting events or generating a sequence of digital states.

In the digital dice circuit, the counter is responsible for generating the six possible dice states. Instead of counting indefinitely, the circuit is designed to return to the initial state after reaching the sixth state.

The counter therefore follows a repeating sequence and provides the digital information required by the decoding logic.

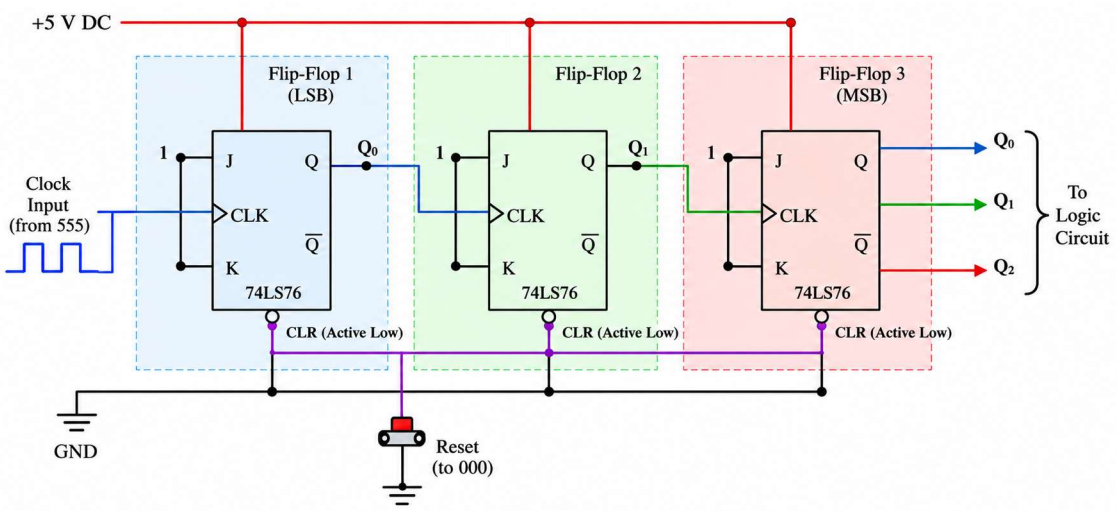


Figure 3.7: Counter Circuit (3-bit Binary Counter using JK Flip-Flops)

3.4.1 Mod-6 Counter

A Mod-6 counter is a counter that has **six valid states**. It counts through six states and then returns to the starting state. This makes it suitable for a digital dice because a conventional dice has six possible outcomes.

A binary counter normally provides more states than required. For example, a 3-bit binary counter can represent eight states from 000 to 111. For the digital dice, only six states are required. Therefore, the unwanted states are prevented or reset so that the counter operates as a Mod-6 counter.

The sequence can be represented as:

000 → 001 → 010 → 011 → 100 → 101 → 000

These six states can be mapped to dice numbers 1 through 6.

Table 3.2: Mod-6 Counter States

Counter State	Binary Output	Dice Number
State 1	000	1
State 2	001	2
State 3	010	3
State 4	011	4
State 5	100	5
State 6	101	6

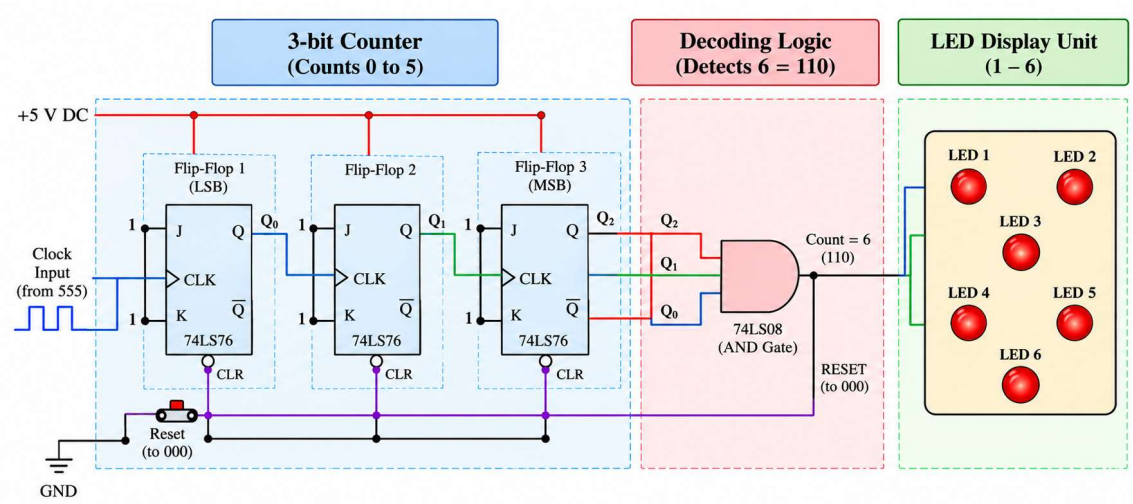


Figure 3.8: Complete Circuit Diagram of LED Dice (Counts 1 to 6)

3.4.2 Counter State Sequence

The counter state sequence determines the order in which the dice numbers are generated. Each clock pulse causes the counter to move to the next state.

When the first state is generated, the logic circuit activates the LED pattern for number 1. The next counter state activates the pattern for number 2, and the process continues until number 6 is reached. After number 6, the counter returns to the first state.

Table 3.3: Counter State Sequence

Clock Pulse	Counter State	Dice Output
1	000	1
2	001	2
3	010	3
4	011	4
5	100	5
6	101	6
7	000	1
8	001	2

This repeating sequence allows the circuit to continuously generate the six possible dice outputs.

3.5 LED Dice Display

The LED dice display is designed to reproduce the appearance of a conventional dice using LEDs. Instead of displaying the number as a digit, individual LEDs are arranged at fixed positions corresponding to the six possible pip positions.

The logic circuit controls these LEDs according to the current counter state. Different combinations of LEDs are switched ON to represent numbers from 1 to 6.

This arrangement provides a simple and visually attractive output and makes the operation of the digital dice easy to demonstrate.

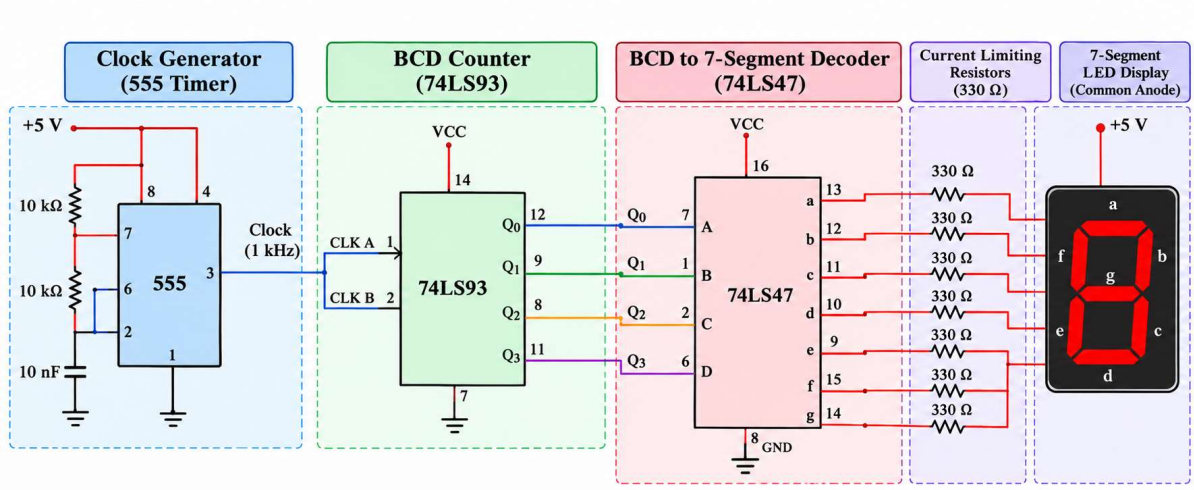


Figure 3.9: 7-Segment LED Display using 74LS93 and 74LS47

3.5.1 LED Arrangement

The LEDs are arranged so that the display resembles the pip pattern of a conventional dice. A seven-position arrangement is sufficient because the centre position is used for the odd numbers, while the remaining positions are shared by the corner and middle LEDs required for the number 6.

The LED positions are identified as P1 to P7. P1 and P2 represent the upper-left and upper-right positions, P3 represents the centre position, P4 and P5 represent the lower-left and lower-right positions, and P6 and P7 represent the middle-left and middle-right positions. The same positions are combined in different ways to obtain all six dice patterns.

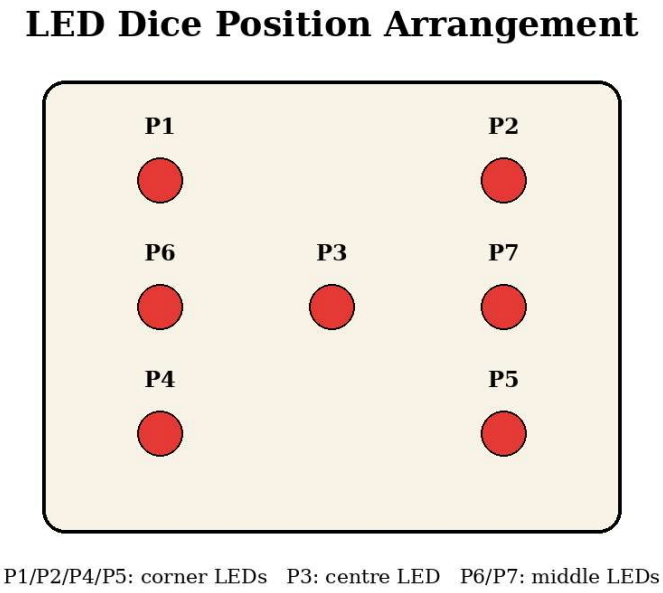


Figure 3.10: LED Position Arrangement

Table 3.4: LED Positions and Functions

LED Position	Location	Purpose
P1	Top-left	Corner pip
P2	Top-right	Corner pip
P3	Centre	Centre pip
P4	Bottom-left	Corner pip
P5	Bottom-right	Corner pip
P6	Middle-left	Used for 6
P7	Middle-right	Used for 6

3.5.2 Dice Number Patterns

The required dice number is represented by switching ON a specific combination of LEDs. The decoding circuit receives the counter state and activates only the LEDs belonging to that number. This approach reduces the complexity of the display because the same LEDs are reused for more than one output pattern.

For number 1, only the centre LED is switched ON. Number 2 uses two diagonally opposite LEDs, while number 3 adds the centre LED. Number 4 uses the four corner LEDs, number 5 adds the centre LED to the four corners, and number 6 uses the two columns of three LEDs.

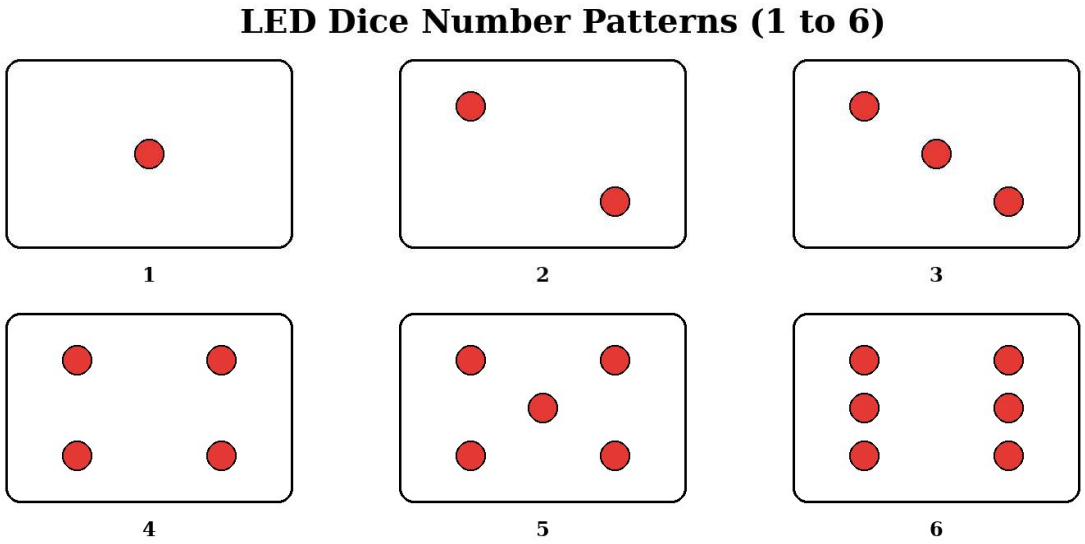


Figure 3.11: Dice Number Patterns from 1 to 6

Table 3.5: Dice Number and LED Pattern

Dice Number	LEDs Switched ON	Pattern
1	P3	Centre
2	P1, P5	Two diagonal positions
3	P1, P3, P5	Three diagonal positions
4	P1, P2, P4, P5	Four corners
5	P1, P2, P3, P4, P5	Four corners + centre
6	P1, P2, P4, P5, P6, P7	Two columns of three

3.6 Power Supply

The power supply provides the required DC voltage to the clock circuit, counter, logic gates and LED display. A regulated supply is preferred because the digital ICs require a stable operating voltage for reliable switching. The supply section can use a transformer, rectifier, filter and voltage regulator to obtain a regulated DC output from an AC source.

For a low-voltage prototype, a regulated +5 V DC source can be used. The clock generator and logic ICs are supplied from this rail, while suitable current-limiting resistors are connected in series with the LEDs. Proper common grounding is maintained throughout the circuit to provide a stable reference for all digital signals.

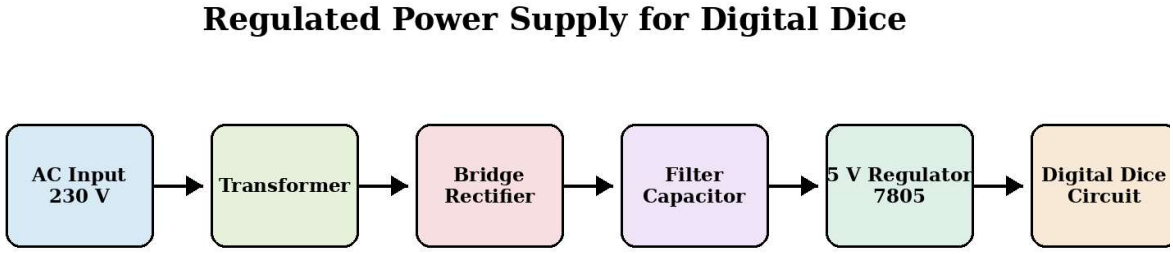


Figure 3.12: Regulated Power Supply for Digital Dice

Table 3.6: Power Supply Requirements

Component	Typical Supply	Purpose
Clock / Pulse Generator	+5 V DC	Generates timing pulses
Counter IC	+5 V DC	Generates digital states
Logic Gate ICs	+5 V DC	Decodes counter outputs
LED Display	+5 V DC	Displays dice pattern
Current-Limiting Resistors	As required	Limits LED current

CHAPTER 4

OUTPUT & WORKING OF DIGITAL DICE

4.1 Number Generation

The digital dice generates its output by converting clock pulses into a repeating sequence of six valid digital states. The clock signal causes the counter to change state, and the logic circuit interprets each state as one dice number. The selected state is then represented by the corresponding LED pattern.

4.1.1 Counter Sequence Generation

A three-bit counter can represent eight binary states, but the digital dice requires only six valid states. The circuit therefore uses a Mod-6 sequence. In the simplified mapping used in this project, the six states 000, 001, 010, 011, 100 and 101 correspond to dice numbers 1 through 6 respectively. After the sixth state, the counter returns to the first state.

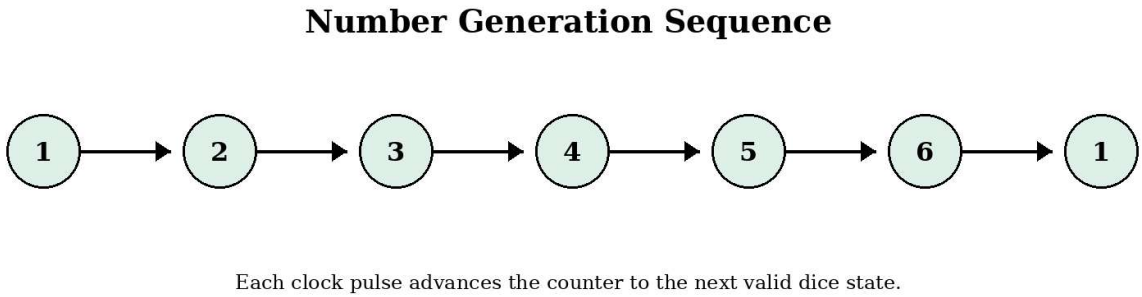


Figure 4.1: Number Generation Sequence of Digital Dice

Table 4.1: Counter Sequence and Dice Output

Clock Pulse	Binary State	Dice Number
1	000	1
2	001	2
3	010	3
4	011	4
5	100	5
6	101	6
7	000	1
8	001	2

4.1.2 Timing and Clock Signal

The clock determines the rate at which the counter changes. A higher clock frequency causes the displayed state to change more rapidly, while a lower frequency makes the sequence easier to observe.

During operation, the user can apply a selection or stop action so that the currently active state is retained as the visible dice result.

4.2 Logic Gate Operation

The logic gates form the decoding section of the system. The counter outputs are combined using AND, OR and NOT operations to identify the required state. When the logic condition for a particular number becomes true, the corresponding LED control signal is activated.

AND gates are useful for detecting a specific combination of binary inputs. NOT gates provide complemented signals when a zero condition is required, while OR gates combine multiple valid conditions. The decoding network therefore converts the binary counter state into individual LED control signals.

Table 4.2: Logic Gate Operation

Gate	Operation	Role in Digital Dice
AND	$Y = A \cdot B$	Detects required state combinations
OR	$Y = A + B$	Combines valid control conditions
NOT	$Y = \overline{A}$	Produces complemented counter signals

4.3 Dice Output Patterns

The output of the decoding circuit is connected to the LED display. Only the LEDs required for the current dice number are switched ON. This provides an immediate visual representation of the generated state and avoids the need for a separate numerical display.

Table 4.3: Dice Output Patterns

Output	LED Pattern	Visual Representation
1	P3	Centre LED
2	P1 + P5	Two diagonal LEDs
3	P1 + P3 + P5	Three diagonal LEDs
4	P1 + P2 + P4 + P5	Four corner LEDs
5	P1 + P2 + P3 + P4 + P5	Five LEDs
6	P1 + P2 + P4 + P5 + P6 + P7	Six LEDs

4.4 Number Decoding

Number decoding is the process of converting the binary state of the counter into the required LED control signals. Each valid counter state is associated with one dice number. The decoder therefore acts as the link between the sequential counter and the visual display.

For example, when the counter produces 010, the decoding logic identifies the third valid state and activates the three-LED pattern. When the counter reaches 101, the sixth pattern is selected and six LEDs are switched ON. The decoding process repeats whenever the counter returns to the first state.

4.5 Reset Operation

Reset operation is used to return the counter to its initial state. A reset input can clear the counter outputs so that the sequence starts again from the first valid state. This is useful during testing, after power-up, and when the user wants to restart the number generation process.

The reset circuit must be connected according to the active-low or active-high reset requirement of the selected counter IC. In the practical circuit, the reset control is also useful for eliminating unwanted states during the Mod-6 operation.

4.6 Practical Output Observation

During practical testing, the clock circuit is first powered and the counter is allowed to run through its sequence. The LEDs should change according to the order 1, 2, 3, 4, 5 and 6. The output can be checked visually by comparing the illuminated LEDs with the expected dice patterns.

If an incorrect LED pattern appears, the counter outputs, logic-gate connections, LED polarity, resistor values and common ground should be checked. The reset operation can then be used to restart the sequence and verify the circuit again.

4.7 Output Characteristics

The main output characteristics of the digital dice are the six discrete display states, the clock-dependent transition rate, and the visible LED representation. The circuit produces digital states rather than a continuous output. The LED brightness depends on the supply voltage, current-limiting resistance and the electrical characteristics of the selected LEDs.

Table 4.4: Output Characteristics of Digital Dice

Characteristic	Expected Behaviour
Number of valid outputs	6
Output range	1 to 6
Counter sequence	Repeating Mod-6 sequence
Display type	LED pip pattern
Control signal	Digital logic levels
Reset function	Returns counter to initial state

CHAPTER 5

CIRCUIT DESIGN & SIMULATION

5.1 Algorithm for Digital Dice

The algorithm describes the sequence of operations performed by the digital dice circuit. It starts by initializing the counter, generating clock pulses, updating the counter state, decoding the state and activating the corresponding LEDs. The process repeats continuously until the system is reset or the selected output is retained.

• Start the digital dice circuit.

• Initialize or reset the counter to the first state.

• Generate continuous clock pulses.

• Update the counter output for every clock pulse.

• Decode the current counter state.

• Switch ON the LEDs corresponding to the decoded dice number.

• Continue the sequence until reset or selection is applied.

• Repeat the process for the next dice cycle.

5.2 Flowchart for Digital Dice

The flowchart represents the sequence of operations performed by the Digital Dice system. It shows how the clock signal is generated, how the counter changes its state, how the logic circuit decodes the counter output, and how the corresponding LED pattern is displayed.

The operation starts with the initialization of the digital dice circuit. A clock or pulse signal is then applied to the counter. The counter generates a sequence of digital states corresponding to the dice numbers from 1 to 6. The logic circuit decodes the generated state and activates the appropriate combination of LEDs.

After displaying the selected dice number, the system checks whether the counter has reached the sixth state. If it has, the counter is reset or returned to the initial state. Otherwise, the counter continues to the next state. This process repeats continuously, allowing the system to generate and display numbers from 1 to 6.

Flowchart Operation

Step 1: Start the Digital Dice system.

Step 2: Initialize the counter and digital circuit.

Step 3: Generate the clock/pulse signal.

Step 4: Apply the clock pulse to the counter and increment its state.

Step 5: Decode the counter output using the logic gates.

Step 6: Activate the corresponding LED pattern for the generated dice number from **1 to 6**.

Step 7: Check whether the counter has reached the sixth state.

Step 8: If the sixth state is reached, reset the counter to the initial state.

Step 9: Continue the clock and counting process to generate the next dice output.

Step 10: Repeat the process continuously.

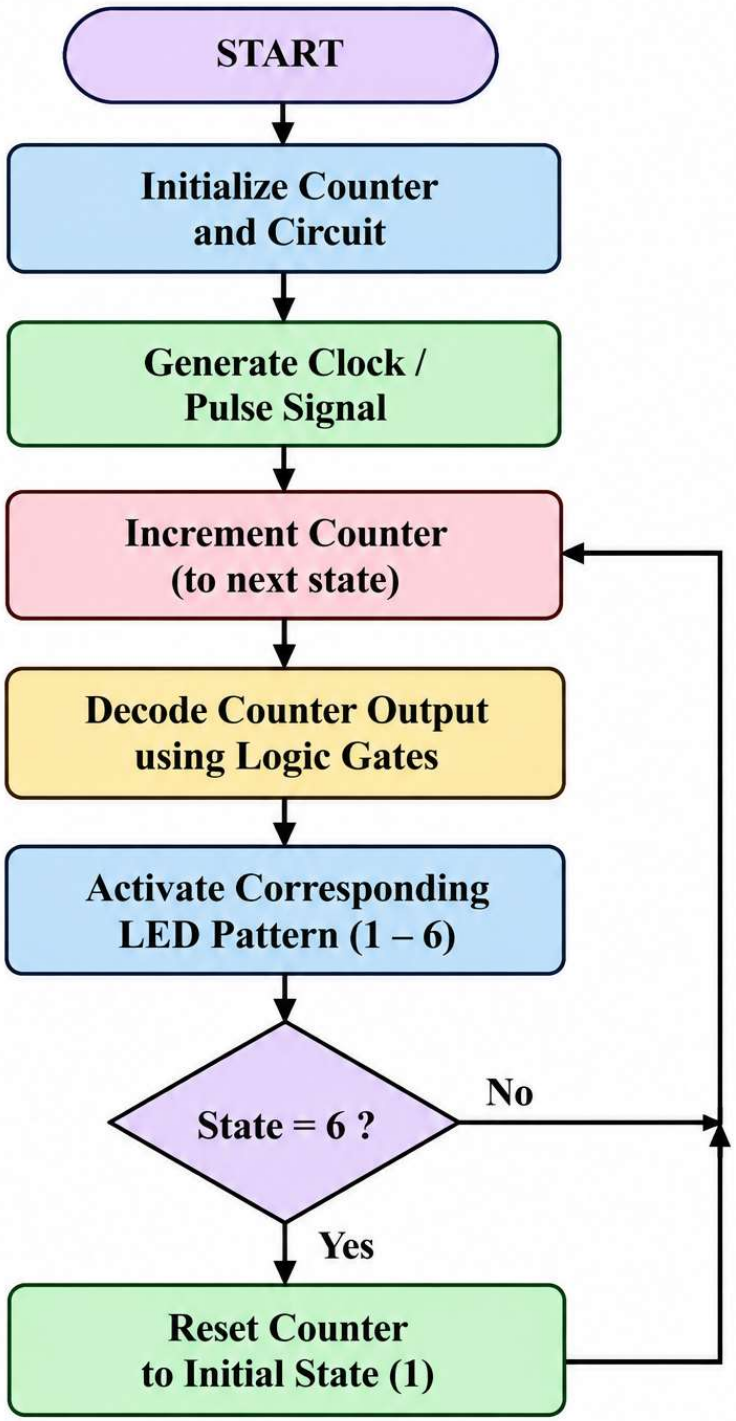


Figure 5.2: Flowchart for Digital Dice

5.3 Logic Circuit Design

The logic circuit is designed by assigning one binary state to each dice number and then generating the required LED control signals for each state. The counter provides the state variables, while the decoding logic determines which LED outputs must be activated for the corresponding dice number.

For the simplified design, the state variables are **Q_2, Q_1 and Q_0**. Since three binary variables can represent eight different combinations, the counter can produce states from **000 to 111**. However, a dice requires only six numbers. Therefore, the valid states are selected from **000 to 101**, corresponding to dice numbers 1 to 6. The remaining combinations, **110 and 111**, are treated as invalid states and can be cleared or prevented using suitable reset logic.

The counter outputs are connected to the logic gates to identify each required state. The decoding circuit combines the counter outputs and their complemented signals using **AND, OR and NOT gates**. Based on the decoded state, the appropriate LED control signals are generated.

Each dice number is represented by a particular combination of LED positions. For example, the state assigned to number 1 activates only the centre LED, while the state assigned to number 4 activates the four corner LEDs. Similarly, different combinations are generated for numbers 2, 3, 5 and 6.

The logic circuit therefore acts as a decoding stage between the counter and the LED display. When a clock pulse changes the counter state, the corresponding logic condition becomes active and the required LED pattern is displayed. After the sixth state, the reset logic returns the counter to the initial state, allowing the sequence to repeat.

This design provides a simple method of converting **binary counter states into visible dice patterns**. It also demonstrates the practical use of combinational logic, sequential counting and LED interfacing in a digital electronic system

5.4 Truth Table

The truth table defines the relationship between the counter state and the visible dice output. It is used to verify that every valid binary state produces only one intended dice pattern.

Table 5.1: Truth Table for Counter State and Dice Output

Q2	Q1	Q0	Dice Output
0	0	0	1
0	0	1	2
0	1	0	3
0	1	1	4
1	0	0	5
1	0	1	6
1	1	0	Invalid / Reset
1	1	1	Invalid / Reset

5.5 Boolean Expression and Simplification

The Boolean expressions for the Digital Dice are obtained from the binary states generated by the counter. Let **Q_2, Q_1 and Q_0** represent the three counter outputs. Each valid counter state is decoded using Boolean logic to identify the corresponding dice number.

Table 5.2: Boolean Minterms for Dice States

Dice Number	Binary State	State Minterm
1	000	$Q2\overline{\ }\,Q1\overline{\ }\,Q0\overline{\ }$
2	001	$Q2\overline{\ }\,Q1\overline{\ }\,Q0$
3	010	$Q2\overline{\ }\,Q1\,Q0\overline{\ }$
4	011	$Q2\overline{\ }\,Q1\,Q0$
5	100	$Q2\,Q1\overline{\ }\,Q0\overline{\ }$
6	101	$Q2\,Q1\overline{\ }\,Q0$

Here, **Q'** represents the complement of the corresponding counter output. These expressions can be implemented using NOT gates to generate the complemented signals and AND gates to detect each individual counter state.

The decoded signals **D_1 to D_6** are then used to control the required LED combinations. For example, when **$D_1 = 1$**, the centre LED is activated to represent dice number 1. When **$D_6 = 1$**, the six required LED positions are activated to represent dice number 6.

The Boolean expressions can be simplified using Boolean algebra or implemented directly using available logic gates. Since only six states are required, the unused states **110 and 111** are treated as invalid states and are handled by the reset logic.

Thus, Boolean simplification reduces unnecessary logic and helps in designing a **simple, reliable and low-cost decoding circuit** for the Digital Dice.

5.6 Circuit Implementation

The Digital Dice circuit can be implemented on a breadboard or a general-purpose prototype board using a **clock/pulse generator, counter circuit, logic-gate ICs, LEDs, current-limiting resistors, and a regulated DC power supply**. The clock generator provides the pulses required to change the counter state, while the counter generates the binary sequence corresponding to the six dice numbers. The outputs of the counter are connected to the logic-gate circuit for decoding. The decoded outputs are then connected to the appropriate **LED positions** to produce the required dice pattern. Each LED is connected through a suitable current-limiting resistor to protect the LED from excessive current.

The counter and logic-gate ICs should be connected according to their respective pin configurations and supply requirements. The power supply and ground connections must be checked carefully before switching ON the circuit. Proper connections are important to avoid incorrect operation or damage to the components.

The reset circuit is used to return the counter to its initial state after the required six-state sequence. The reset operation should be verified separately to ensure that the counter correctly returns to the starting state.

After completing all connections, the circuit can be powered and tested by observing the LED output for each counter state. The LEDs should display the corresponding dice patterns from **1 to 6**, and the sequence should repeat continuously after the sixth state.

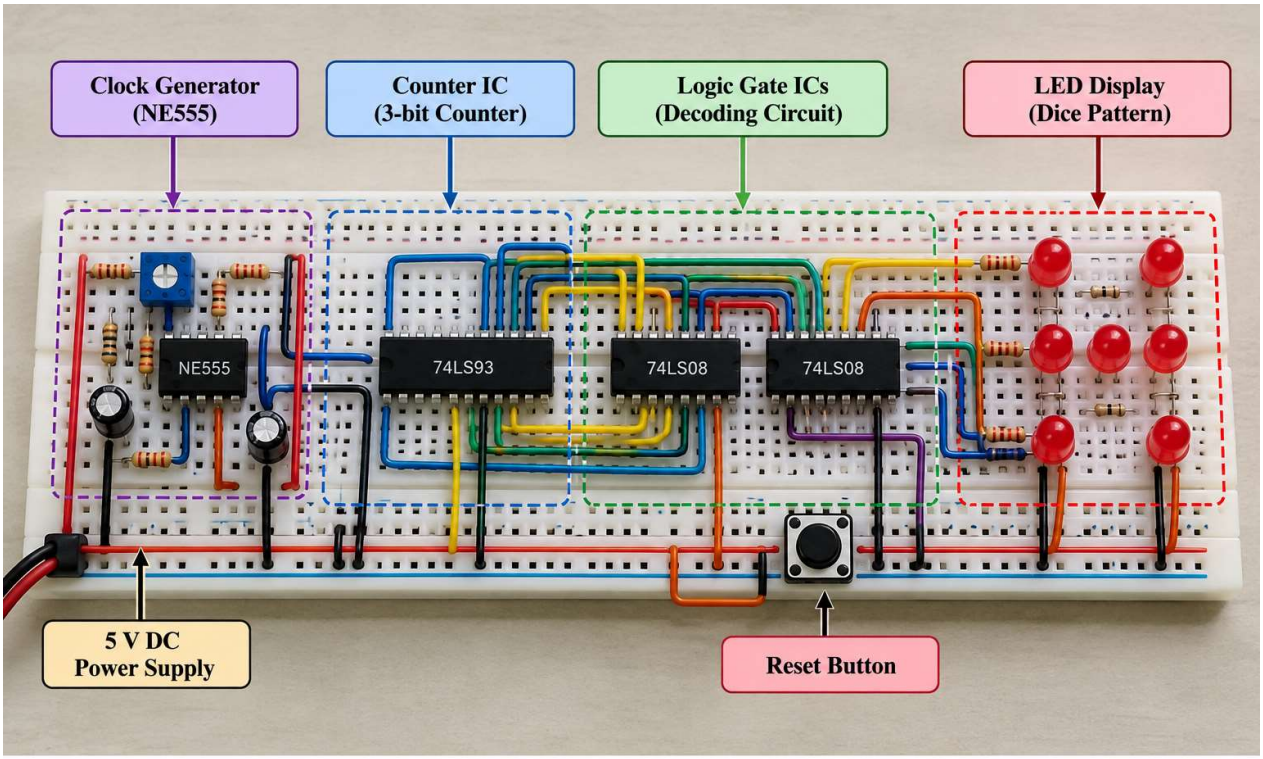


Figure 5.3: Hardware Implementation of Digital Dice

5.7 Simulation Model & Circuit Diagram

Simulation provides a convenient method of checking the digital dice before hardware implementation. The clock, counter, decoding logic and LED display can be represented using equivalent digital components in software such as Logisim, Logisim Evolution or Proteus. The simulation model allows the state transitions and output patterns to be verified without risking hardware components.

In the simulation model, the **clock or pulse generator** is connected to the counter circuit. The clock produces continuous digital pulses that cause the counter output to change from one state to the next. For the digital dice, the counter is designed to operate through six valid states corresponding to the dice numbers 1 to 6. The counter outputs are represented by the binary variables **Q_2, Q_1 and Q_0**.

The outputs of the counter are connected to the **logic decoding circuit**. The decoding circuit identifies the current binary state and produces the required control signals for the LED display.

Each valid counter state corresponds to one dice number. For example, the states 000, 001, 010, 011, 100 and 101 can be assigned to dice numbers 1, 2, 3, 4, 5 and 6 respectively.

The decoded signals are then connected to the **LED display section**. The LEDs are arranged in the form of a conventional dice pattern. According to the counter state, the required combination of LEDs is switched ON to display the corresponding number. Therefore, when the counter changes from one state to another, the displayed dice pattern also changes.

A **reset circuit** is included in the simulation to ensure that the counter does not continue into unwanted states. When the sixth valid state is completed, the counter is returned to the initial state. This produces a repeating sequence of six dice outputs. The reset operation also helps in checking whether the circuit correctly handles invalid counter states.

The circuit diagram in the simulation consists of four major sections: **clock generator, counter, logic decoder and LED display**. These sections are interconnected in the same manner as the proposed hardware circuit. The simulation therefore provides a logical representation of the complete digital dice system before it is physically constructed.

The simulation also helps to identify incorrect connections, logic errors and improper state transitions at an early stage. By observing the output LEDs for each counter state, the designer can verify that the required dice patterns are produced correctly. Once the simulated circuit gives the expected sequence from 1 to 6, the same logic can be implemented using physical ICs, LEDs and other hardware components.

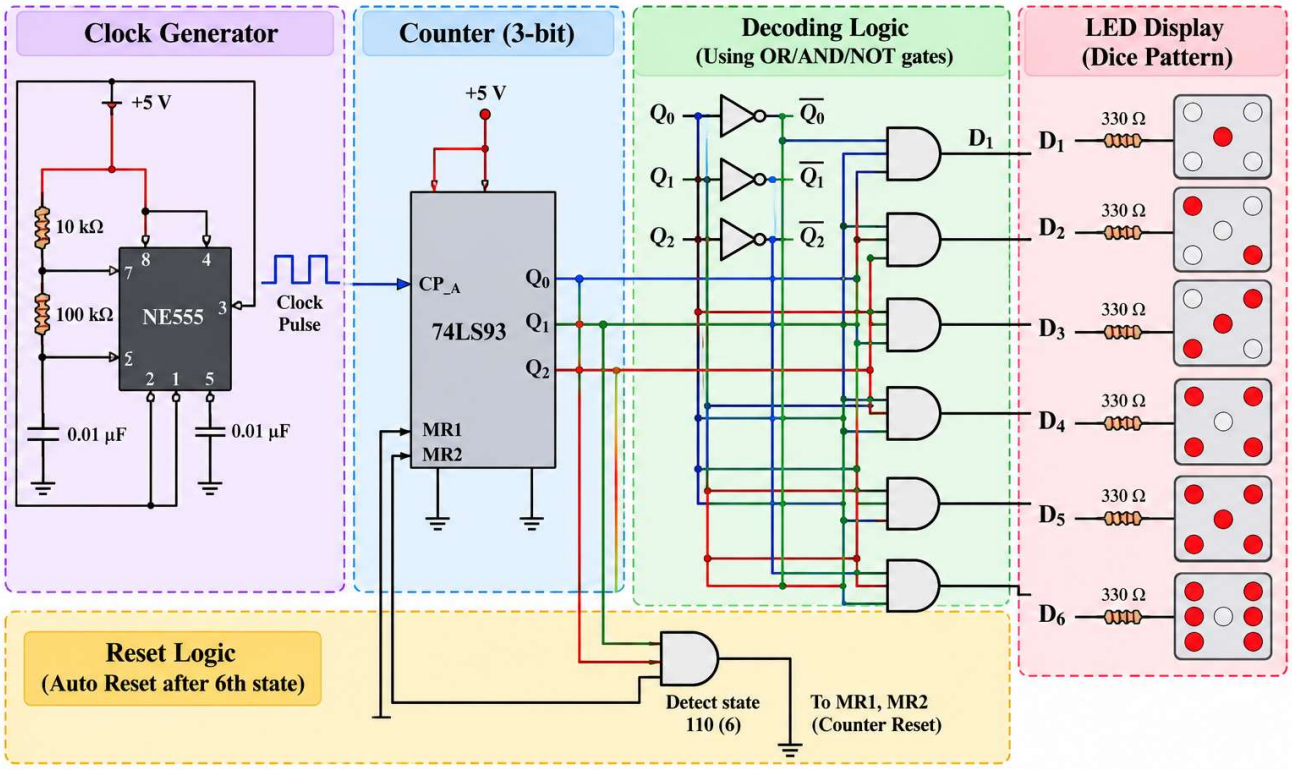


Figure 5.4: Simulation Model and Circuit Diagram of Digital Dice

5.8 Simulation Waveforms

The simulated waveforms show the **clock signal and the corresponding counter outputs**. The clock waveform consists of a continuous sequence of pulses that controls the operation of the counter. Each active clock pulse causes the counter to advance to the next state.

The counter outputs **Q_0, Q_1 and Q_2** change according to the binary counting sequence. The valid states are represented as **000, 001, 010, 011, 100 and 101**, corresponding to dice numbers **1 to 6**. By observing these signals together, the correct state transitions can be verified.

The waveform also shows the relationship between the clock pulses and the counter outputs. Q_0 changes more frequently, while Q_1 and Q_2 change at lower frequencies according to the binary counting operation. This confirms that the counter is responding correctly to the applied clock signal. When the counter reaches the sixth valid state, the reset logic brings the counter back to the initial state. The waveform therefore repeats the six-state sequence continuously. This ensures that invalid states beyond the required dice range are avoided.

The decoded outputs can also be observed along with the counter signals to verify that the appropriate LED pattern is activated for each state. For example, when the counter represents state **000**, the output corresponding to dice number 1 is activated, while state **101** activates the output corresponding to dice number 6.

Thus, the simulation waveform provides a clear method for verifying the **timing, counting sequence, reset operation and dice output generation**. Successful waveform verification indicates that the designed logic circuit is functioning as expected before proceeding to practical hardware implementation.

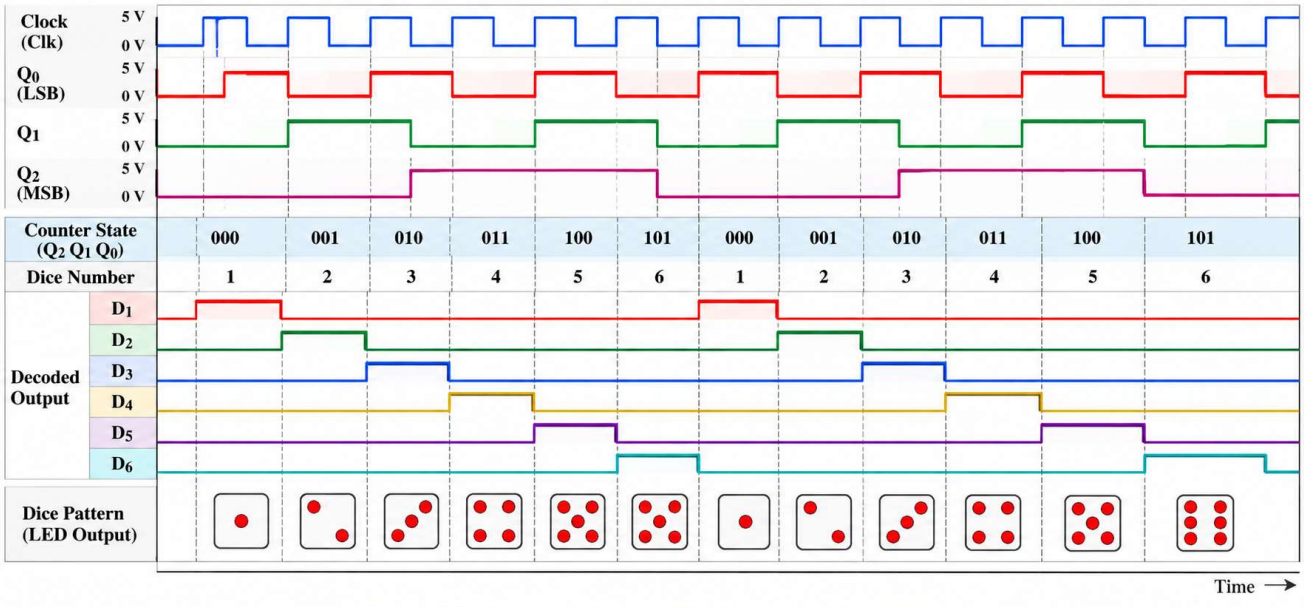


Figure 5.5: Simulated Counter Waveforms for Digital Dice

5.9 Simulation Procedure

The simulation procedure is carried out systematically to verify the operation of the digital dice circuit before hardware implementation. The following steps are followed:

1. **Create the Clock/Pulse Generator:**

First, place a clock or pulse generator in the simulation software. Set a suitable frequency so that the counter can change states clearly. The clock provides the continuous pulses required to operate the counter.

2. **Connect the Counter Circuit:**

Connect the output of the clock generator to the clock input of the counter. Configure the counter to produce the required **Mod-6 sequence**. The counter should generate the binary states **000, 001, 010, 011, 100 and 101**.

3. **Verify Counter Outputs:**

Connect monitoring probes or logic indicators to **Q_2, Q_1 and Q_0**. Run the simulation briefly and check whether the counter changes through the six expected states in the correct order.

4. **Connect the Decoding Logic:**

Connect Q_2, Q_1 and Q_0 to the required **AND, OR and NOT gates**. The decoding circuit identifies each counter state and generates the corresponding output signal for the particular dice number.

5. **Connect the LED Display:**

Connect the outputs of the decoding circuit to the LED dice display. Suitable **current-limiting resistors** are connected in series with the LEDs to prevent excessive current and ensure proper LED operation.

6. **Provide Power and Ground Connections:**

Connect the required DC supply voltage to all the digital components. Ensure that all components share a **common ground** and that the power connections are correctly made before starting the simulation.

7. **Configure the Reset Circuit:**

Connect the reset logic to the counter so that unwanted states are avoided. The reset operation should return the counter to the initial state after completing the required six-state sequence.

8. **Run the Simulation:**

Start the simulation and observe the clock pulses, counter outputs and decoded outputs. The counter should progress through the six valid states according to the applied clock signal.

9. **Observe the LED Patterns:**

Check the LED display for each counter state. The display should produce the corresponding dice patterns for **1, 2, 3, 4, 5 and 6**.

10. **Verify the Complete Operation:**

Compare the simulated output with the expected dice sequence. Check the state transitions, LED patterns and reset operation. If any incorrect output or unwanted state is observed, the circuit connections and logic conditions are checked and corrected.

11. **Record the Simulation Results:**

Finally, record the observed counter states, output patterns and waveforms. The verified simulation results can then be used as a reference for implementing the digital dice circuit using physical hardware components.

5.10 Simulation Results

The simulation results are used to verify the complete operation of the Digital Dice circuit. The clock signal, counter states, decoding logic and LED output are observed during the simulation. The expected output is a continuous sequence of six valid dice states, after which the circuit returns to the initial state.

During the simulation, each clock pulse causes the counter to advance to the next binary state. The valid counter states are **000, 001, 010, 011, 100 and 101**, which are assigned to dice numbers **1, 2, 3, 4, 5 and 6**, respectively. The decoding circuit converts these binary states into the corresponding LED control signals.

The LED display is checked for every dice number. For example, state **000** activates the centre LED to represent number 1, while state **011** activates the four corner LEDs to represent number 4. Similarly, state **101** produces the six-LED pattern representing number 6.

The reset operation is also verified during the simulation. When the reset condition is applied, the counter returns to its initial state. This prevents unwanted counter states and allows the six-state sequence to repeat correctly.

Table 5.3: Simulation Test Results

Test Case	Expected Result	Observation
Clock applied	Counter advances	Verified in simulation
State 000	Dice number 1	Centre LED pattern
State 011	Dice number 4	Four corner LEDs
State 101	Dice number 6	Six-LED pattern
Reset applied	Initial state restored	Counter returns to starting state

From the simulation, the **clock, counter, decoding circuit, LED display and reset operation** were found to work according to the designed sequence. The waveform observation also confirmed that the counter outputs changed correctly with the applied clock pulses.

The successful simulation provides a basis for proceeding with the **practical hardware implementation** of the circuit.

CHAPTER 6

FUTURE ENHANCEMENT

6.1 Proposed System Enhancements

The present digital dice demonstrates the basic operation of clock generation, counting, logic decoding and LED display. The system can be extended with additional control, display and communication features while retaining the same fundamental concept of generating one of six valid outcomes.

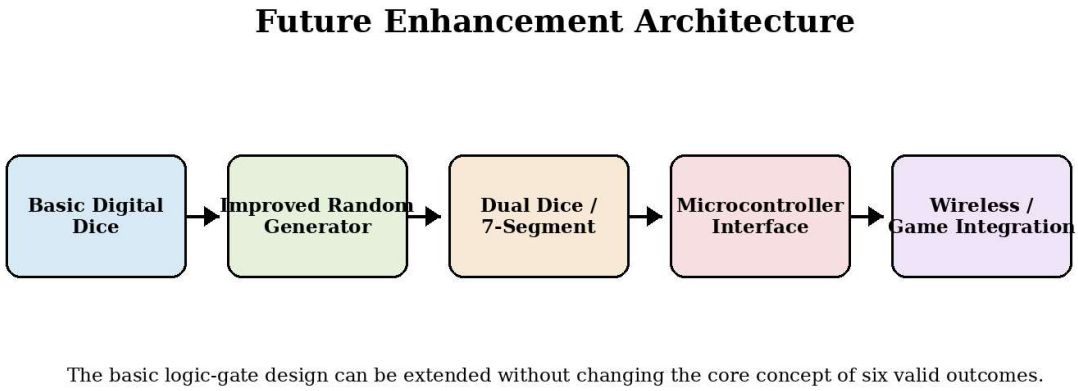


Figure 6.1: Future Enhancement Architecture

6.2 Improved Random Number Generation

A simple counter produces a repeating sequence and therefore does not provide true random behaviour by itself. A more advanced design can use a pseudo-random sequence generator such as a linear feedback shift register (LFSR), or sample a rapidly changing digital state using a user-triggered selection signal. This can make the dice behaviour less predictable while maintaining six valid outputs.

6.3 Dual Dice System

The circuit can be expanded to operate two dice simultaneously. A second counter and decoding section can be added so that two independent outputs are displayed. The two results can be used for board games, probability experiments and digital gaming demonstrations.

6.4 Seven-Segment Display Enhancement

A seven-segment display can be added as an alternative to the pip-style LED display. In this arrangement, the generated dice number is shown as a decimal digit from 1 to 6. The seven-segment display can improve readability when the circuit is used in laboratory demonstrations or electronic game interfaces.

6.5 Microcontroller-Based Digital Dice

The logic-gate implementation can be replaced or supplemented by a microcontroller. A microcontroller can generate the timing, perform state selection, control the LEDs and implement more advanced randomization algorithms. This approach also makes it easier to add switches, displays, sound and data communication.

6.6 Electronic Game Integration

The digital dice can be integrated into electronic board games and educational gaming systems. The output can be connected to a controller that moves a player position, updates a score or selects an action according to the generated number.

6.7 Wireless/IoT-Based Enhancement

Wireless communication can be added to transmit the generated dice value to another device. A microcontroller with a suitable wireless interface can send the result to a computer or mobile application for remote gaming, data logging or classroom demonstrations.

6.8 Overall Improvement of the System

Future versions can combine improved randomization, dual-dice operation, multiple display options, user controls and wireless communication. These improvements can transform the basic educational circuit into a more interactive digital system while retaining the six-state counting concept.

Table 6.1: Proposed Future Enhancements

Enhancement	Main Benefit
Improved random generation	Less predictable output
Dual dice	Two simultaneous results
Seven-segment display	Direct numerical display
Microcontroller control	Flexible programming and control
Game integration	Automatic electronic gameplay
Wireless interface	Remote result transmission

6.9 Advantages

- Simple digital circuit concept and easy to understand.
- Demonstrates both sequential and combinational logic.
- Provides a clear LED-based visual output.
- Low-cost components can be used for a prototype.
- Suitable for laboratory demonstration and digital electronics learning.
- Can be expanded with displays, microcontrollers and wireless features.

6.10 Applications

• Digital electronics laboratory demonstrations.

• Educational kits for learning counters and logic gates.

• Electronic board-game prototypes.

• Probability and counting experiments.

• Interactive entertainment systems.

• Prototype input for simple gaming applications.

6.11 Limitations

• A basic counter produces a deterministic repeating sequence.

• The LED display requires multiple output lines and current-limiting resistors.

• Incorrect logic connections can produce unwanted LED patterns.

• The basic design does not provide true randomness without an additional randomization method.

• The number of outputs is limited to six by the Mod-6 design.

• Practical hardware operation depends on stable power and correct component selection.

Table 6.2: Limitations of the Basic Digital Dice

Parameter	Limitation
Randomness	Basic counter sequence is deterministic
Output range	Limited to six dice values
Hardware	Requires several ICs and LEDs
Power	Requires stable regulated supply
Display	Multiple LEDs require several control lines

CHAPTER 7

CONCLUSION

The Digital Dice Using Logic Gates project demonstrates the practical application of fundamental digital electronics concepts in a simple electronic system. The circuit combines a clock or pulse generator, counter, logic-gate decoding circuit and LED display to generate and represent six valid dice outcomes.

The counter provides the sequence of digital states, while the logic gates decode the states and control the appropriate LED combinations. The Mod-6 operation ensures that the system uses only the six required dice outputs. The LED arrangement provides a clear visual representation of the generated number and makes the circuit suitable for demonstration and educational purposes.

The project also provides practical understanding of sequential circuits, combinational logic, state generation, decoding, LED interfacing and reset operation. Simulation can be used to verify the logic before hardware implementation, reducing wiring errors and improving confidence in the final prototype.

Although the basic counter-based system is deterministic, it provides a strong foundation for future development. Improved random-number generation, dual-dice operation, seven-segment displays, microcontroller control, electronic game integration and wireless communication can be incorporated to create more advanced versions of the system.

Thus, the Digital Dice Using Logic Gates serves as a compact and practical mini project for demonstrating how basic digital components can be combined to form a complete interactive electronic application.